

Selection of MPPT Chip

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Due to chip shortage due to the COVID 19 Pandemic we have to have many options regarding the chips we are using in our system. Currently in the beginning of 2022 we are facing same issue and we want to have an option to the MPPT chip that we are currently using. We currently have SPV1040 in our EPS board in NEPALPQ-1: PocketQube Training Kit.

Objective of the study

- To find a chip that can work as an option to the SPV1040
- It should be easily available in aliexpress/ arrow/ or digikey/ mouser.

Main features of SPV1040 that is required from other chips

- 0.3 V to 5.5 V operating input voltage
- uses embedded hardware MPPT to charge the battery

Some additional features of SPV1040 that we like

- 95% efficient
- Charging can be shut down
- Reverse polarity protection in the input side
- Simple circuit and does not contain many components
- Easy to solder because of its TSSOP8 package.

Some considerations

- Due to the chip shortage, we might have to go to the extent that we have to select a boost converter that boosts the voltage of approx 2V to the required voltage and a Li-ion battery charger that charges the battery.

Comparative Table

Parameters	SPV1040	LTC3105	SPV1050	ADP5090
Input Voltage Range [V]	0.3-5.5	0.22-5	0.45-5.5 *	0.1-3.6
Package	TSSOP8	Yes	VFQFPN [3X3]	16-Lead LFCSP
Uses MPPT?	Yes	Yes	Yes	Yes
Price[\$]	3.5 Euro [Mouser]	6.5 Euro	3.9\$ [digikey]	5.88 Euro [mouser]

Parameters	SPV1040	LTC3105	SPV1050	ADP5090
Availability **	available in mouser	available in mouser, aliexpress	aliexpress [expensive]	available in mouser, arrow

Some other options include ADP5091, ADP5092. ***

- ** As of Jan 2022

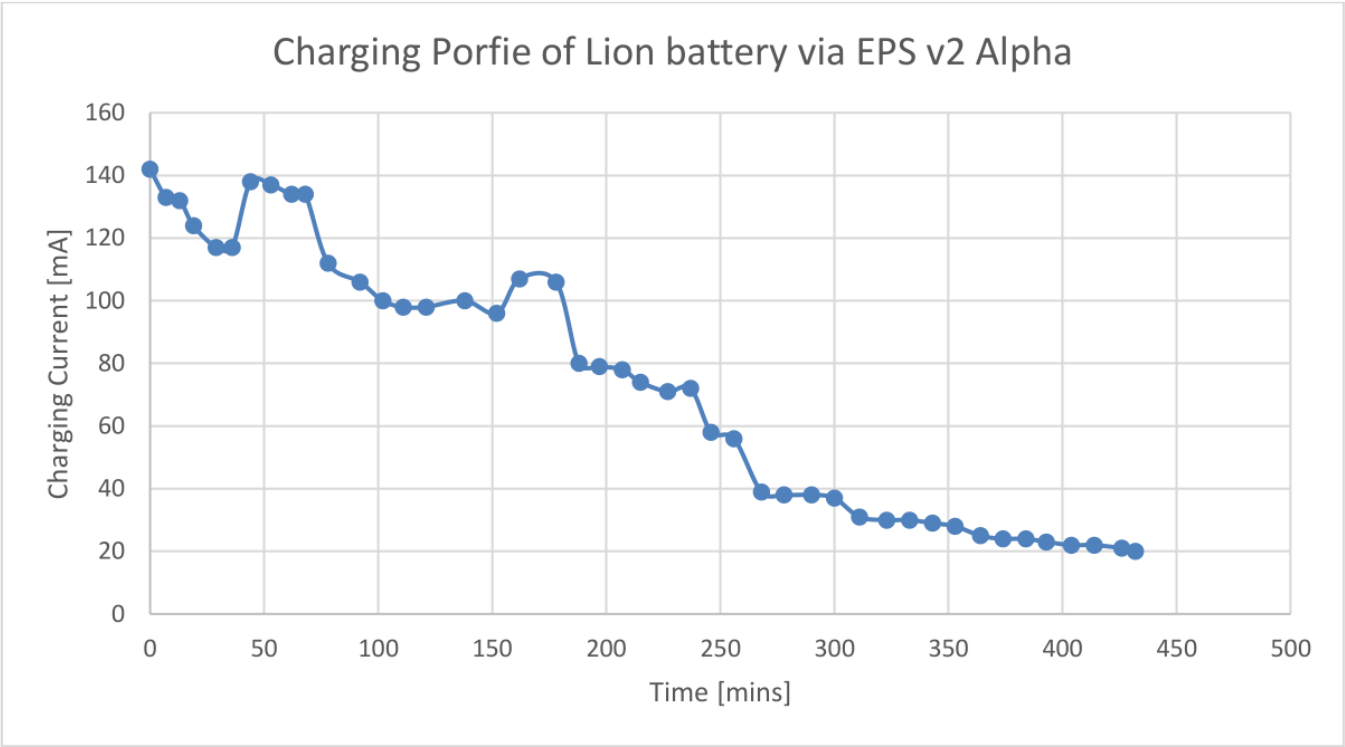
Some Preliminary Studies and Designs

1. LTC3105

In this, we desgined a board that can be suitable for our purose for EPS and evaluated the evaluation board that we desgined.

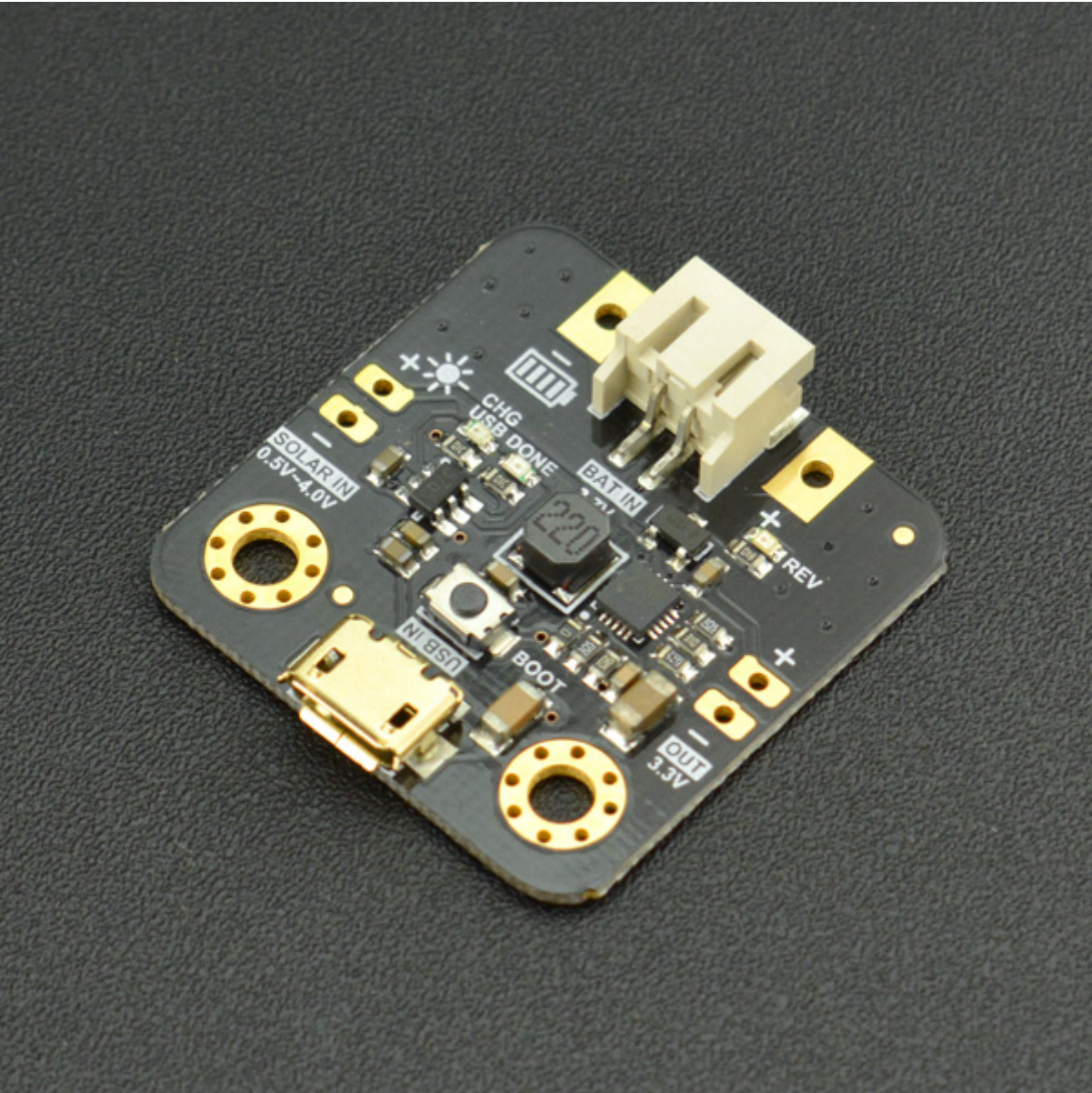


The board has been tested and the results are as follows:



2. SPV1050

In this we ordered a board from dfrobot that uses SPV105 and evaluated that board.



Boost Converters

To convert the 2V from the solar panel we can also use boost converter. Here we try to find the high efficient boost converter which can be paired with a battery charger that charges the battery in the EPS board.

Parameters	L6920
Input Voltage Range [V]	0.6-5.5
Package	TSSOP8
Price[\$]	1.8\$ [aliexpress]
Availability **	available in aliexpress
Additional Features	Shut Down Pin, Reverse input voltage protection, low voltage detection at input, 3.3v and 5V fixed or adjustable output voltage [2-5.2V]

• ** As of Jan 2022

While using boost converter we also need to use battery chargers in parallel to connect the voltage to the battery.

Battery Charger

This chip will charge a 1-cell Li-ion battery and stop after the charging has been completed.

Parameters	MCP73831T
Input Voltage Range [V]	3.75-6
Output Voltage [V]	4.2, 4.35, 4.4, 4.5
Package	SOT-23-5, DFN
Price[\$]	0.2\$ [aliexpress]
Availability **	available in aliexpress

- ** As of Jan 2022

Recomendation

1. Option-1: Use LTC3105
2. Option-2: Use L6920 and MCP73831T

References

- SPV1040 Datasheet
- *<https://www.st.com/en/power-management/energy-harvesting-and-solar-charging-ics.html#overview>
- *** <https://www.analog.com/en/parametricsearch/11503#/p5573=20m|1&p5574=500m|5>